Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Real Book Exercises, Lecture 4

How to use this list. Every exercise below is a real exercise from one of the course books, reproduced verbatim: the source line gives author, book, chapter, exercise, the printed page and, in brackets, the page your PDF viewer shows for the file in `Livros/`, and whether the source gives the answer. Muñoz-Garcia & Toro-Gonzalez print a worked answer right after each exercise; MWG answers are in the MWG solutions manual in `Livros/`; Tadelis, Gibbons and Cabral give no answer in the book. When an exercise refers to a model, game or figure stated elsewhere in its book, that material is copied into the same block under "Referenced material", so every exercise can be solved where it is printed. There is one block per lecture of the real schedule. Each block has ten exercises in strictly rising difficulty, rated from 1 (apply a definition) to 5 (graduate level) next to each heading: three foundations, five exam level, two advanced. Try each exercise in the space provided before opening the book's answer.

Difficulty path. Lecture 4: from definitions (1) through entry, Stackelberg, hold-up and the legal-system game (Tadelis 8.7) to MWG 9.B.5 (5), a five-part general-payoff argument on strategy counts, multiple SPNE and when moving first guarantees the simultaneous-move payoff, with counterexamples.

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L4	1 (31)	Gibbons 2.12†	135 [142]	Found.	Strategy, subgame and SPE in a repeated game	1	none
	2 (32)	Tadelis 7.6‡	148–149 [167–168]	Found.	Entry game: payoffs from preferences, all Nash equilibria	1	none
	3 (33)	Tadelis 8.8	171–172 [190–191]	Found.	Entry deterrence: a non-credible threat	2	none
	4 (34)	Tadelis 8.6†	170–171 [189–190]	Exam	Sunk investment before Cournot; threshold F^*	3	none
	5 (35)	Muñoz-Garcia Ch. 4 Ex. 11‡	180 [192]	Exam	Stackelberg, asymmetric costs; vs. monopoly, Cournot, Bertrand	3	book
	6 (36)	Tadelis 8.11‡	172–173 [191–192]	Exam	Sequential vetoes; Nash equilibria that are not SPE; removing an option	3	none
	7 (37)	Tadelis 8.4†	170 [189]	Exam	Leader, two followers: subgames; a Nash equilibrium that is not SPE	4	none
	8 (38)	Tadelis 8.13	173–174 [192–193]	Exam	Hold-up; a posted price as commitment	4	none
	9 (39)	Tadelis 8.7‡	171 [190]	Adv.	Credibility of suing; design of the legal system	4	none
	10 (40)	MWG 9.B.5†	302 [162]	Adv.	Observation, multiple SPNE, when moving first pays	5	manual

#: item number in this lecture's list, then its number in the combined list (`lista-reais-lectures-01-05.pdf`) in brackets. Items not in the list for Lectures 1–4 (`lista-reais-lectures-01-04.tex`) are marked: † added in the 30-item version, ‡ added in the 50-item version. Every other item is reused from that list with its statement unchanged, some in a different block or position. "Diff.": difficulty from 1 (apply a definition) to 5 (graduate level). Pages: printed page, then the PDF viewer page in brackets (offsets in `Livros/book-registry.json`; MWG is a two-pages-per-sheet scan). "Answer": *book* = worked answer printed after the exercise; *manual* = MWG solutions manual in `Livros/`; *none* = no answer in the book. Books: Tadelis (2013), *Game Theory: An Introduction*; Gibbons (1992), *Game Theory for Applied Economists*; Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed.; Mas-Colell, Whinston & Green (1995), *Microeconomic Theory* (MWG); Cabral (2017), *Introduction to Industrial Organization*, 2nd ed.

Block 4 — Credible threats and subgame perfection *(Lecture 4)*

Exercise 1. (item 31 of the combined list) *Repeated-game definitions* [Foundations; difficulty 1/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.3), Problem 2.12, p. 135 [PDF 142]. No answer in the book. Reproduced verbatim.

2.12. What is a strategy in a repeated game? What is a subgame in a repeated game? What is a subgame-perfect Nash equilibrium?

Resolution:

Exercise 2. (item 32 of the combined list) *Entering an Industry* [Foundations; difficulty 1/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 7 "Preliminaries", Exercise 7.6 "Entering an Industry", pp. 148–149 [PDF 167–168]. No answer in the book. Reproduced verbatim.

7.6 Entering an Industry: A firm (player 1) is considering entering an established industry with one incumbent firm (player 2). Player 1 must choose whether or not to enter the industry. If player 1 enters the industry then player 2 can either accommodate the entry or fight the entry by waging a price war. Player 1's most-preferred outcome is entering with player 2 not fighting, and its least-preferred outcome is entering with player 2 fighting. Player 2's most-preferred outcome is player 1 not entering, and its least-preferred outcome is player 1 entering with player 2 fighting.

a. Model this as an extensive-form game tree (choose payoffs that represent the preferences).

b. How many pure strategies does each player have?

c. Find all the Nash equilibria of this game.

Resolution:

Exercise 3. (item 33 of the combined list) *Entry Deterrence 1* [Foundations; difficulty 2/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.8 "Entry Deterrence 1", pp. 171–172 [PDF 190–191]. No answer in the book. Reproduced verbatim.

8.8 Entry Deterrence 1: NSG Corporation is considering entry into the local phone market in the San Francisco Bay Area. The incumbent, S&P Corporation, predicts that a price war will result if NSG enters. If NSG stays out, S&P earns monopoly profits valued at \$10 million (net present value of profits) while NSG earns zero. If NSG enters, it must incur irreversible entry costs of \$2 million. If there is a price war, each firm earns \$1 million. S&P always has the option of accommodating entry (i.e., not starting a price war). In such a case both firms earn \$4 million (net present value). Suppose that the timing is such that NSG first has to choose whether or not to enter the market. Then S&P decides whether to "accommodate entry" or "engage in a price war." What is the subgame-perfect equilibrium outcome to this sequential game? Set up a game tree.

Resolution:

Exercise 4. (item 34 of the combined list) *Investment in the Future* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.6 "Investment in the Future", pp. 170–171 [PDF 189–190]. No answer in the book. Reproduced verbatim.

8.6 Investment in the Future: Consider two firms that play a Cournot competition game with demand $p = 100 - q$ and costs for each firm given by $c_i(q_i) = 10q_i$. Imagine that before the two firms play the Cournot game firm 1 can invest in cost reduction. If it invests the costs of firm 1 will drop to $c_1(q_1) = 5q_1$. The cost of investment is $F > 0$. Firm 2 does not have this investment opportunity.

a. Find the value F^* for which the unique subgame-perfect equilibrium involves firm 1 investing.

b. Assume that $F > F^*$. Find a Nash equilibrium of the game that is not subgame perfect.

Resolution:

Exercise 5. (item 35 of the combined list) *Stackelberg with Two Firms* [Exam level; difficulty 3/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 4 "Sequential-Move Games with Complete Information", Exercise 11 "Stackelberg with Two Firms"[A], p. 180 [PDF 192]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only).

Consider a leader and a follower in a Stackelberg game of quantity competition. Firms face an inverse demand curve $p(Q) = 1 - Q$, where $Q = q_L + q_F$ denotes aggregate output. The leader faces a constant marginal cost $c_L > 0$ while the follower's constant marginal cost is $c_F > 0$, where $1 > c_F \geq c_L$, indicating that the leader has access to a more efficient technology than the follower.

Part (a) Find the follower's best response function, BRF_F, i.e., $q_F(q_L)$.

Part (b) Determine each firm's output strategy in the subgame perfect equilibrium (SPNE) of this sequential-move game.

Part (c) Under which conditions on c_L can you guarantee that both firms produce strictly positive output levels?

Part (d) Assuming cost symmetry, i.e., $c_L = c_F = c$, determine the aggregate equilibrium output level in the Stackelberg game, and its associated equilibrium price. Then compare them with those arising in (1) a monopoly; in (2) the Cournot game of quantity competition; and (3) in the Bertrand game of price competition (which, under cost symmetry, is analogous to a perfectly competitive industry) [*Hint*: For a more direct and visual comparison, depict the inverse demand curve $p(Q) = 1 - Q$, and locate each of these four aggregate output levels in the horizontal axis (with its associated prices in the vertical axis).].

Resolution:

Exercise 6. (item 36 of the combined list) *Resident Assistant Selection with a Twist* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.11 "Resident Assistant Selection with a Twist", pp. 172–173 [PDF 191–192]. No answer in the book. Reproduced verbatim.

8.11 Resident Assistant Selection with a Twist: Two staff managers in the ΠΒΦ sorority, the house manager (player 1) and kitchen manager (player 2), must select a resident assistant from a pool of three candidates: $\{a, b, c\}$. Player 1 prefers a to b, and b to c. Player 2 prefers b to a, and a to c. The process that is imposed on them is as follows: First, the house manager vetoes one of the candidates and announces the veto to the central office for staff selection and to the kitchen manager. Next the kitchen manager vetoes one of the remaining two candidates and announces it to the central office. Finally the director of the central office assigns the remaining candidate to be a resident assistant at ΠΒΦ.

a. Model this as an extensive-form game (using a game tree) in which a player's most-preferred candidate gives a payoff of 2, the second most-preferred candidate gives a payoff of 1, and the last candidate gives 0.

b. Find the subgame-perfect equilibrium for this game. Is it unique?

c. Are there Nash equilibria that are not subgame-perfect equilibria?

d. Now assume that before the two players play the game, player 2 can send an alienating e-mail to one of the candidates, which would result in that candidate withdrawing her application. Would player 2 choose to do this, and if so, with which candidate?

Resolution:

Exercise 7. (item 37 of the combined list) *Industry Leader* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.4 "Industry Leader", p. 170 [PDF 189]. No answer in the book. Reproduced verbatim. (Gibbons Problem 2.6, p. 133, poses the same game and asks only for the subgame-perfect outcome; it is not repeated in this list.)

8.4 Industry Leader: Three oligopolists operate in a market with inverse demand given by $P(Q) = a - Q$, where $Q = q_1 + q_2 + q_3$, and q_i is the quantity produced by firm i. Each firm has a constant marginal cost of production, c, and no fixed cost. The firms choose their quantities dynamically as follows: Firm 1, which is the industry leader, chooses $q_1 \geq 0$. Firms 2 and 3 observe q_1 and then simultaneously choose q_2 and q_3, respectively.

a. How many proper subgames does this dynamic game have? Explain briefly.

b. Is it a game of perfect or imperfect information? Explain briefly.

c. What is the subgame-perfect equilibrium of this game? Show that it is unique.

d. Find a Nash equilibrium that is not a subgame-perfect equilibrium.

Resolution:

Exercise 8. (item 38 of the combined list) *Junk Mail Advertising* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.13 "Junk Mail Advertising", pp. 173–174 [PDF 192–193]. No answer in the book. Reproduced verbatim.

8.13 Junk Mail Advertising: Suppose a single good is owned by a single seller who values it at $c > 0$ (he can consume the good and get a payoff of c). There is a single buyer who has a small transportation cost $k > 0$ to get to and back from the seller's store, and he values the good at $v > c + k$. The buyer first decides whether to make the commute or stay at home, not buy the good, and receive a payoff of 0. If the buyer commutes to the store, the seller can then make the buyer a take-it-or-leave-it price offer $p \geq 0$. The buyer can then accept the offer, pay p, and get the good, or he can walk out and not buy the good. Assume that c, v, and k are common knowledge.

a. As best you can, draw the extensive form of this game. What is the best response of the buyer at the node at which he decides whether to accept or reject the seller's offer?

b. Find the subgame-perfect equilibrium of the game and show that it is unique. Is it Pareto optimal?

c. Are there Nash equilibria that yield a higher payoff to both players as compared to the subgame-perfect equilibrium you found in (b)?

d. Now assume that before the game is played the seller can, at a small cost $\varepsilon < (v - c - k)$, send the buyer a postcard that commits the seller to a certain price at which the buyer can buy the good (e.g., "bring this coupon and get the good at a price p"). Would the seller choose to do so? Justify your answer with an equilibrium analysis.

Resolution:

Exercise 9. (item 39 of the combined list) *Debt and Repayment* [Advanced; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 8 "Credibility and Sequential Rationality", Exercise 8.7 "Debt and Repayment", p. 171 [PDF 190]. No answer in the book. Reproduced verbatim.

8.7 Debt and Repayment: A project costing \$100 yields a gross return of \$110. A lender (player 1) is approached by a debtor (player 2) requesting a standard loan contract to complete the project. If the lender chooses *not to offer* a loan, then both parties earn nothing. If the lender chooses to *offer* a loan of \$100, the debtor can realize the project's gains and is obliged by contract to repay \$105. For simplicity assume that money is continuous and that the debtor can choose to return any amount of money $x \leq 110$. Ignore the time value of money. Assume first that no legal system is in place that can compel the debtor to repay, so that default on the loan (less than full repayment) carries no repercussions for the debtor.

a. Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?

b. Now assume that there is a legal system in place that allows the lender to *voluntarily* choose whether or not to sue when the debtor defaults and repays an amount $x < 105$. Furthermore assume that it is *costless* to use the legal system (it is supplied by the state), and if the lender sues a debtor that defaulted, the lender will get the \$105 repaid in full. After paying the lender, the borrower will pay a fine of \$5 to the court, above and beyond the repayment. Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?

c. Are there Nash equilibria in the game described in (b) that are not subgame-perfect equilibria?

d. Now assume that using the legal system is *costly:* if the lender sues, he pays lawyers a *legal fee* of \$105 (this is the lawyers' price; it is unrelated to the contract). The rest of the game proceeds as before. (If the lender sues a debtor that defaulted, the lender will get repaid in full; after paying the lender, the borrower will pay a fine of \$5 above and beyond the repayment.) Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?

e. Are there Nash equilibria in the game described in (d) that are not subgame-perfect equilibria?

f. Now assume that a change in the law is proposed: upon default, if a debtor is sued he has to first repay the lender \$105 and then pay the legal fees of \$105 above and beyond repayment of

the loan, and no extra fine is imposed. Should the lender be willing to pay to have this change in the law enacted? If so, how much?

g. If you were a social planner, would you implement the suggested law?

Resolution:

Exercise 10. (item 40 of the combined list) *Observation, multiple SPNE and moving first*

[Advanced; difficulty 5/5]

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 9 "Dynamic Games", Exercises, Exercise 9.B.5^{B} (E. Maskin), p. 302 [PDF 162, left half]. Answer in the MWG solutions manual (`Livros/`; 9.B.5 is there). Reproduced verbatim; the label of the last part is blurred in the scan and is read as (e), as in the solutions manual.

9.B.5^{B} (E. Maskin) Consider a game with two players, player 1 and player 2, in which each player i can choose an action from a finite set M_i that contains m_i actions. Player i's payoff if the action choices are (m_1, m_2) is $\phi_i(m_1, m_2)$.

(a) Suppose, first, that the two players move simultaneously. How many strategies does each player have?

(b) Now suppose that player 1 moves first and that player 2 observes player 1's move before choosing her move. How many strategies does each player have?

(c) Suppose that the game in (b) has multiple SPNEs. Show that if this is the case, then there exist two pairs of moves (m_1, m_2) and (m_1', m_2') (where either $m_1 \neq m_1'$ or $m_2 \neq m_2'$) such that either

$$\text{(i)} \quad \phi_1(m_1, m_2) = \phi_1(m_1', m_2')$$

or

$$\text{(ii)} \quad \phi_2(m_1, m_2) = \phi_2(m_1', m_2').$$

(d) Suppose that for any two pairs of moves (m_1, m_2) and (m_1', m_2') such that $m_1 \neq m_1'$ or $m_2 \neq m_2'$, condition (ii) is violated (i.e., player 2 is never indifferent between pairs of moves). Suppose also that there exists a pure strategy Nash equilibrium in the game in **(a)** in which π_1 is player 1's payoff. Show that in any SPNE of the game in **(b)**, player 1's payoff is at least π_1. Would this conclusion necessarily hold for any *Nash* equilibrium of the game in **(b)**?

(e) Show by example that the conclusion in **(d)** may fail either if condition (ii) holds for some strategy pairs (m_1, m_2), (m_1', m_2') with $m_1 \neq m_1'$ or $m_2 \neq m_2'$ or if we replace the phrase *pure strategy Nash equilibrium* with the phrase *mixed strategy Nash equilibrium.*

Resolution: